November 2026

Cryptography | Group 4

SECURE FILE SYSTEM

D4 - Midterm Presentation

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SYSTEM OVERVIEW

What it does

Protects a document before it leaves the sender's trusted device, so it can be stored or sent anywhere and only authorized recipients can recover it.

Problem it solves

Networks and storage cannot be trusted: an attacker can read, copy, modify, replay or delete files. Security must not depend on them.

Confidentiality

Integrity

Sender authenticity

Recipent authorization

ARCHITECTURE

TRUSTED: Sender

Vault Sender App
Encrypts and signs the file

Key Store
Sender private key, recipient public keys

Trust Boundary 1

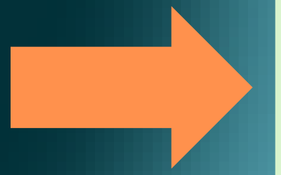


UNTRUSTED: Transfer

Encrypted Container
Ciphertext, bound metadata, signature

Network / Storage
Mallory can copy, modify, replay, delete

Trust Boundary 2



TRUSTED: Recipient

Vault Recipient App
Validates, authorizes, decrypts

Key Store
Recipient private key, sender public key

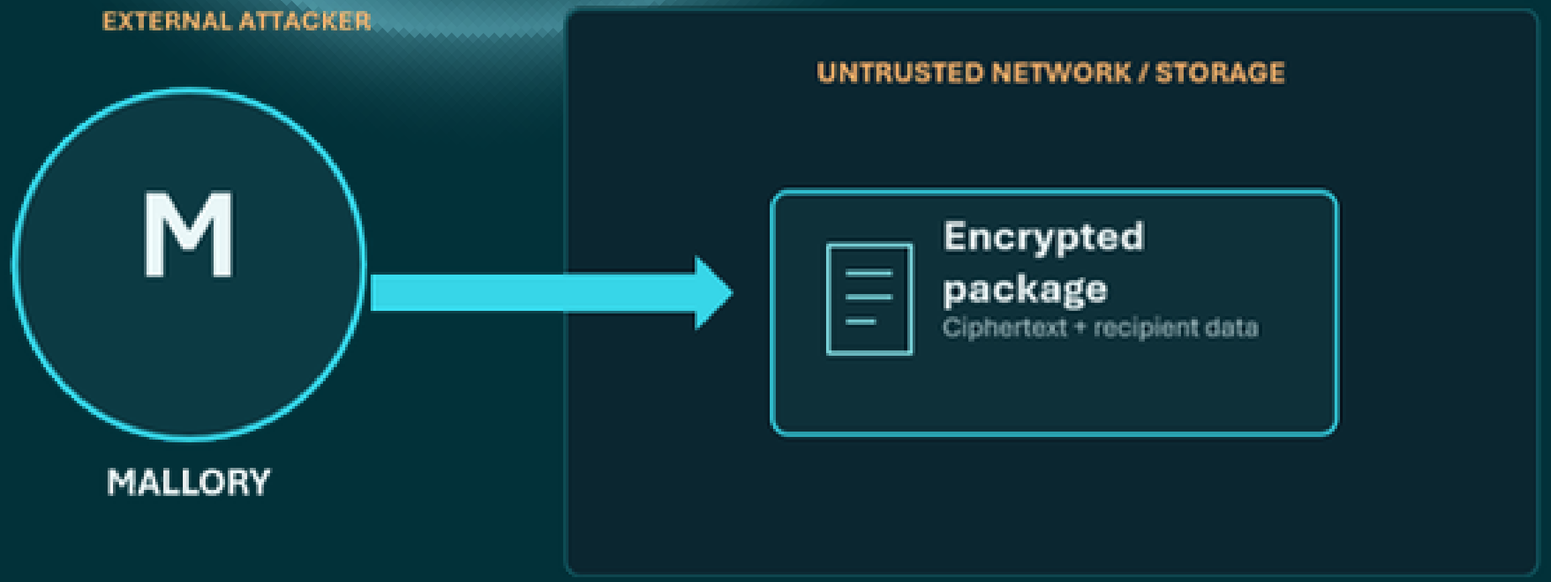
1. Encrypt and sing

2. Share (Untrusted)

3. Verify and decrypt

Threat Model: Mallory

The transfer channel and storage are untrusted.



Mallory can:

01 **Observe / copy**

02 Modify

03 Replay

04 Delete

Protected Assets & Trust Assumptions

Security holds when trusted endpoints and recipient keys are handled safely.

WHAT WE PROTECT

01 **Original documents**

02 Encrypted package and ciphertext

03 Recipient list and metadata

04 File keys and private keys

WHAT WE ASSUME

- Sender and recipient devices are trusted
- Private keys remain protected
- Recipient public keys are verified by Key ID over a separate trusted channel
- Network and storage may be observed or changed

Data Tier

CRYPTOGRAPHIC DESIGN

D2 **Cipehrtext Gen.**

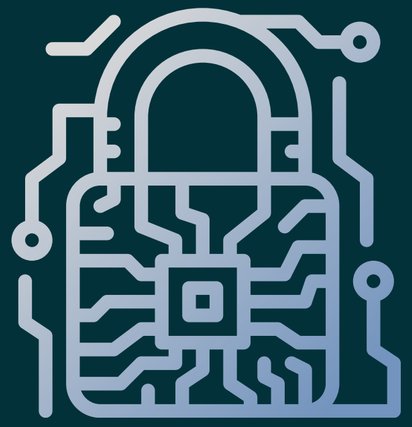


- AEAD: AES-256-GCM. 256b Key size & 128b tag
- decrypt_and_verify: No plain plaintext given unless tag is verified.
- 12B Nonce
- The key is derivated from the passphrase (human or not)
- Freshly generated keys every time.
- AAD Headers are visible but authenticated

Flow is “Passphare → KEK → DEK → AES-GCM, and the header is authenticated”

CRYPTOGRAPHIC DESIGN





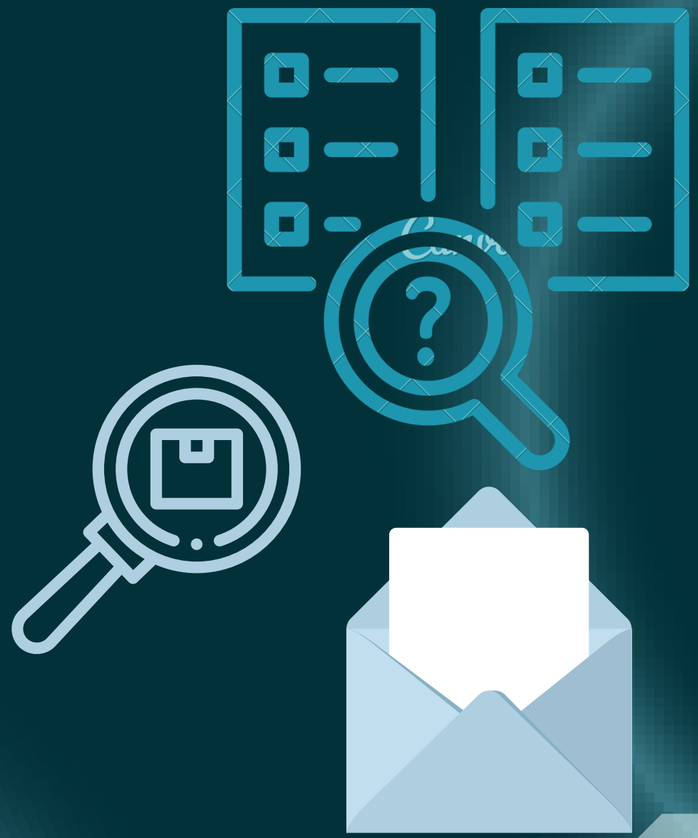
D2 **Hybrid Encription**

- The same file is secured for multple recipients.
- A single key file, but 1 key pair per user (with Ed25519)
- Only PKs are shared
- Works over D2: Passphrase replaced by random 256b key.
- Extra AAD: Outer header includes the list, payload is D2 header
- Any tamper invalidates the acces for everyone
- Identifies the key, not the user

Flow is *“One key per file → ciphered per user → validate the list”*



CRYPTOGRAPHIC DESIGN



D3 **Decription - Outer Layer**

- Validates the strcuture before touching the keys
- Were you invited?
- ECIES gets the encrypted key file

D2 **Decription - Inner layer**

- Use the key file to recover KEK and DEK
- Verify the GCM tag
- Recreate the plain text (original file)

`Flow is “Check the entry → recover the key → verify the tag → write the file”`

DATA STATE	ENCRYPTION STANDARD	KEY MANAGEMENT
Data at Rest	AES-256	AWS KMS / Azure Key Vault
Data in Transit	TLS 1.3	Automated certificate rotation
Data in Use	Confidential Computing	Enclave-based processing

Privacy Controls

- Data classification (Public, Internal, Confidential, Restricted)
- DLP policies for email, cloud storage, endpoints
- GDPR, CCPA, HIPAA compliance where applicable
- Data retention & secure deletion policies

LIVE DEMO

FUTURE IMPROVEMENTS

FROM D5 & D6

- Digital Signatures
- User Authentication
- Key Managment
- Key Lifecycle Definition

Estimated progress is 50% donoe.

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THANK YOU

Securing Innovation, Protecting Trust

Questions, suggestions or
complaints - please don’t